

Nonlinear element

W21 (2021) — English translation

The electrical circuit consists of a constant voltage source U_0 , a capacitor C , a nonlinear element НЭ and a switch K . The current-voltage characteristic of the nonlinear element is as follows:

$$\left(\frac{I}{I_0}\right)^2 + \left(\frac{U}{U_0}\right)^2 = 1$$

при $I > 0$ и $I = 0$ при $|U| \geq U_0$ (см. рис.). Величина I_0 известна. В начальный момент напряжение на конденсаторе равно $U_0/2$. Ключ K close.

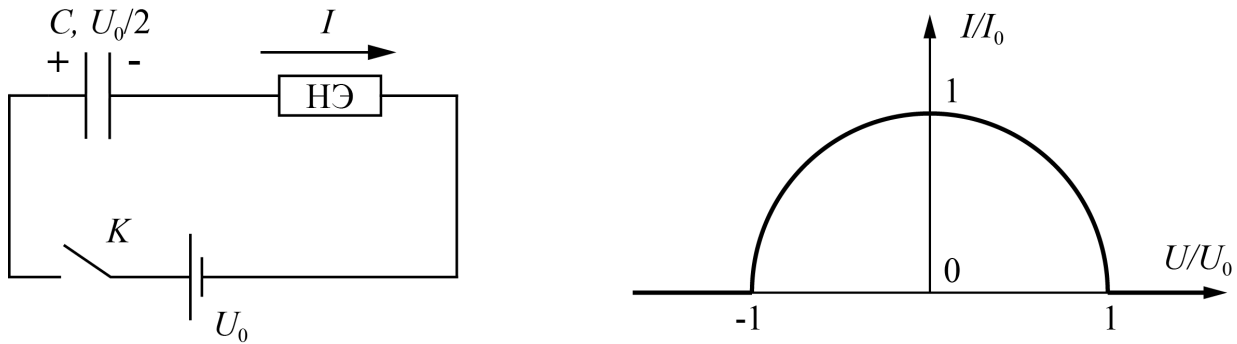


Fig. 1.

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|-----------|---|-------------|
| A1 | Find the maximum power P_1 released on the nonlinear element. | 2.00 |
| A2 | Find the maximum rate of change of energy of the capacitor P_2 and the time after closing the switch Δt after which it is achieved. | 8.00 |

Source: <https://pho.rs/p/80>